

Question justifications

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1 Pre-session Questionnaire

1.1 Participant Experience Information

The first few questions of this survey are used to confirm the identity of the participant being interviewed.

The next few questions examine the participant's experiences. '5 years' is a conservative amount of time that was chosen, because technology is ever-changing, and is projected to change in the near future (<https://www.hunimed.eu/news/10-ways-technology-is-changing-healthcare/>). In addition, the level of care that was received previously may have been very different than what is practiced today (<https://prod.mgma.com/articles/from-error-to-excellence-redesigning-care-whole-system-quality>).

Experience with a personal orthopaedic surgery has a minor overlap with having a knee surgery due to their similar nature. In that sense, an orthopaedic patient may be more empathetic to those that have had surgeries recently. They may also impart their first-hand understanding of such an experience and provide richer, more nuanced insights into the potential benefit and use of the application.

1.2 Participant device literacy

In order to properly operate the application, there is a minimum level of device literacy needed. These questions specifically target smartphone users. Some populations may not have equal access to the level of care (socioeconomic reasons) (<https://www.cmaj.ca/content/185/6/E263>) that is assumed for this application's use, and that is somewhat portrayed through the use of these questions. The purpose of these questions asserts the need for further exploration if/when the target audience is widened and should further development of the application occurs.

1.3 Participant perspective on privacy

This section outlines some private aspects involved in using the app. These statements should be confirmed with participants to ensure that a well-informed decision to participate and use it has been made.

2 Session questionnaire

2.1 Create Account

The account creation functionality is a similar one to that of existing apps. It uses a conventional flow that should be familiar to users. The ease of this process is evaluated. Once a participant has created an account, they are directed

to the Homescreen. This page has different capabilities based on the type of user.

2.2 Homescreen

On a patient's screen, it displays a history of completed exercises and their dates. On a PT's screen, it displays the names of their patients. Once they have selected a patient, the screen is populated with the patient's activity. Both the patient's and PT's homescreen only serve as a record, displaying patient activity history. Since both patient and PTs pages look very similar, the main questions posed for this section are UI perspectives. Drawing on both perspectives, these questions serve to validate the homescreen's minimal aesthetic, confirming that it contains only essential information and is not visually cluttered. The same logic follows the progress tab.

2.3 Exercise

Despite the UI being almost the same, there is a functional difference when the two types of users interact with the Exercises page which is addressed by the questions.

In the case of the patient, the questions aim to determine whether the elements are presented clearly and in a straightforward manner. The example video under the complete exercise instructions was implemented at a later stage, necessitating investigation into whether it was a helpful decision.

In the PT's case, an additional functionality is introduced. While the exercise list maintains the same appearance, clicking on the exercise reveals a text box which allows modifications to be made to the instructions. The intuitiveness and ease needed to be ascertained.

2.4 Record

PTs are unable to access this function and the page remains inaccessible. This functionality is not intended for them and was not developed further.